Knowing the Graph: Does Relational Structure Improve Linear-Probe Detection of Related AI-Safety Concepts?

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Second-model results: peer replication and extension by **p0ss** (HatCat)

Abstract

AI-safety concepts such as manipulation, deception and sycophancy are related to one another: they form a graph of siblings and hubs rather than a list of independent categories. We ask whether knowing that graph improves linear-probe detection of 15 related concepts in the residual stream of Gemma-2-9B, with a peer replication on Gemma-4-E4B. Across four iterations, with every powered test after the first pre-registered, we find two results. First, relations help as *local hard negatives*: a concept's declared siblings are more confusable with it than unrelated concepts (ΔF1 $+0.055$ across 15 concepts, Wilcoxon $p = 1.5 \times 10^{-4}$, pre-registered), and training a concept's probe against its direct neighbours improves discrimination by about $+0.06$ F1 on both models. Exploratory controls suggest roughly a third of the sibling effect is shared vocabulary. Second, we find no measurable benefit from the graph's *connectivity*. When the evaluated sibling is withheld from the negatives, the remaining graph structure recovers $\Delta\text{F1} \approx +0.01$, and a pre-registered tenfold data-volume sweep does not raise it: at 70 passages per concept per context the mean is $+0.005$ (95% CI $[-0.005, +0.014]$), and net of vocabulary overlap it is -0.006. This rules out a connectivity benefit much larger than $+0.015$ F1 on this graph, but not one near our minimum detectable effect of 0.014. The test is also limited by the graph itself (15 concepts, 26 edges, at most four alternate paths per edge), so we scope the null to this graph. We describe a post-hoc account, *independent-constraint boundary coverage*, that fits both results, and draw a deployment consequence: each negative-selection scheme seals only the boundary face it trains against. On a pre-specified test with legitimate-influence passages (negotiation, advertising, lobbying), none of the three schemes tested reduces false positives, which stay near 0.23 on held-out contexts.

1 Introduction

Deployable monitoring for AI-safety-relevant behaviour increasingly relies on concept-detection probes: linear read-outs of a model's internal activations that flag, for example, deception or sycophancy (Alain and Bengio, 2016; Zou et al., 2023; MacDiarmid et al., 2024; Goldowsky-Dill et al., 2025). Safety concepts rarely come alone. Manipulation borders persuasion; misinformation and disinformation differ only in intent; sycophancy, unwarranted praise and excessive agreement form a tight family (Carroll et al., 2023; Park et al., 2024b; Sharma et al., 2024). A monitor needs one detector per concept across a whole family, and the family's structure is often available for free in an ontology. The obvious hope is that this structure can substitute for data.

We test a specific form of that hope, inherited from the HatCat concept-lens programme (p0ss, 2026). It has two parts. (H1) Relationship-aware training improves discrimination between related concepts. (H2) The benefit scales with graph connectivity: if two concepts are joined by many

vertex-disjoint paths (Menger, 1927), a probe should be able to triangulate their boundary from the surrounding relationships even without direct evidence about that boundary.

Our answer, after four iterations, is bounded. H1 holds in a narrow form: relations are useful as *hard negatives*, and the benefit is local to the boundary being tested. H2 finds no support at the data volumes and graph we could test. We also report where the useful part stops: negatives chosen from the graph do nothing for legitimate influence that sits just outside the concept family.

The route to this answer is part of the result. Our first iteration produced an apparent positive that did not survive a powered replication (Appendix A). After it, every powered test was pre-registered with a locked decision rule, every positive result was checked by an independent refutation pass, and the study was replicated and extended on a second model by p0ss, who originated the hypothesis. The pre-registration documents and their commit history are public (Appendix B).

Contributions.

1. A pre-registered test of whether declared relations identify the confusable pairs, with a placebo-graph control and an exploratory vocabulary control that together bound the effect (§4).
2. A pre-registered within-model data-volume sweep of the held-out-edge test, the condition that separates graph connectivity from adjacency, with across-edge confidence intervals, a reported minimum detectable effect, and a vocabulary-adjusted estimate (§5).
3. Cross-model agreement: Gemma-2-9B (this work) and Gemma-4-E4B (peer replication) give the same pattern of effects on independently generated datasets (§4, §5).
4. An account of the pattern in terms of independent boundary constraints rather than path counts, stated with its limits (§6).
5. A measured deployment boundary: a rejection test on legitimate influence, pre-specified before it was run, in which none of the three negative-selection schemes tested helps (§7).

2 Related work

Probing methodology. Linear probes read out information that is linearly decodable from a layer (Alain and Bengio, 2016; Kim et al., 2018). Decodability is not the same as use, and a probe can succeed on surface regularities in the data rather than the intended property (Belinkov, 2022). Control tasks (Hewitt and Liang, 2019) address this by comparing against a probe's performance on a task it should fail. Our out-of-distribution context split and vocabulary regressions serve the same purpose: lexical overlap is the strongest competing explanation for every effect we report.

Concept geometry. Under the linear representation hypothesis, concepts correspond to directions (Park et al., 2024a). Park et al. (2025) extend this to categories and hierarchies, finding that hierarchically related WordNet concepts occupy orthogonal subspaces in Gemma and LLaMA-3. Our question is complementary. We take a curated, non-hierarchical graph of safety concepts and ask whether its relations are *useful for training probes*, not whether they are reflected in geometry. Our geometry check (§6) finds local adjacency reflected and graded graph distance not. Related work on truth directions (Marks and Tegmark, 2023; Burns et al., 2023) and representation engineering (Zou et al., 2023) motivates reading safety-relevant properties linearly. Raw cosine similarity between transformer activations is dominated by a common direction (Ethayarajh, 2019), and mean-centring is the standard correction (Mu and Viswanath, 2018); we apply it.

Probes for safety-relevant behaviour. Linear probes have been used to detect strategic deception (Goldowsky-Dill et al., 2025) and backdoored "sleeper agent" behaviour (MacDiarmid et al., 2024). Both report strong separation and both note that false positives on benign but related inputs are the practical bottleneck. Our near-out-of-set test (§7) measures exactly that failure for a family of influence concepts.

Hard negatives. Choosing negatives near the decision boundary is a long-standing technique in metric and contrastive learning (Schroff et al., 2015; Kalantidis et al., 2020; Robinson et al., 2021). Our positive result is an instance of it: graph siblings are good hard negatives. The open question we test is whether a graph adds anything *beyond* supplying nearby negatives.

Rejection. A classifier without a reject option assigns every input to some class (Chow, 1970; Geifman and El-Yaniv, 2017). Confidence thresholds are a weak out-of-distribution detector (Hendrycks and Gimpel, 2017); training against auxiliary outliers is stronger (Hendrycks et al., 2019). Our two-level probe and "other" tier (§7) are applications of these ideas to concept lenses.

Pre-registration. Pre-registration separates confirmatory from exploratory analysis (Nosek et al., 2018). We follow that distinction throughout and label every analysis that was not locked in advance. A null under a "confidence interval spans zero" rule is weaker than an equivalence test (Lakens, 2017); we report what our intervals exclude rather than claiming equivalence.

3 Setup

Concepts. SET G is 15 AI-safety concepts from the manipulation, deception and influence family (Appendix E), selected from HatCat applied melds layered on the SUMO/WordNet concept pack (p0ss, 2026; Niles and Pease, 2001; Fellbaum, 1998).

Three graphs. The study uses three operationalisations of "related", and we keep them separate. (i) *Ontology graph* (23 edges), derived from HatCat relations and used for the iteration-2 connectivity test (§5); `ManipulativeCommunication` is adjacent to 12 of the other 14 concepts, and 84 of the 105 concept pairs have vertex connectivity $k = 1$. (ii) *Declared siblings* (18 pairs), hand-written discriminating pairs used for the Path-2 hard-negative test (§4). (iii) *Contrastive boundary graph* `SET_G_N15_graph_contrastive_boundary_v1` (26 edges, each with a written rationale and boundary question), used for the held-out-edge test and the scaling sweep. Its hub is again `ManipulativeCommunication`, with degree 7.

Data. Each concept has a one-sentence definition. Passages were generated by Claude Opus 4.8 (Anthropic, 2026) in four surface-distinct contexts (workplace, online, personal relationships, marketplace). For the scaling sweep we generated 70 passages per concept per context (4,200 total). A blind audit asked the same model to assign each passage to one of the 15 concepts from definitions alone: mean self-match was 94.7%, lowest for the hub concept `ManipulativeCommunication` (70.4%). The near-out-of-set controls (§7) are 84 passages for three legitimate-influence concepts, all 84 self-matched under the same audit.

Probes and evaluation. We mean-pool the residual stream at layer 18 (Gemma-2-9B, $d = 3584$, Gemma Team, 2024; Gemma-4-E4B, $d = 2560$, the same block index, Google DeepMind, 2026) and fit balanced logistic-regression probes. Evaluation is always out of distribution: probes are trained

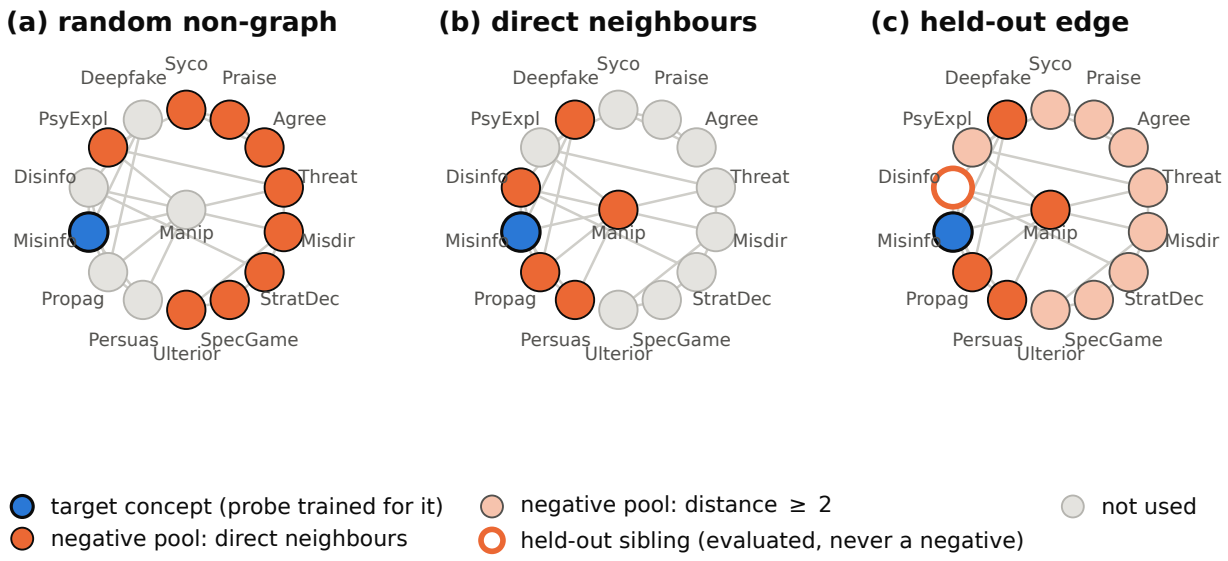


Figure 1: Negative-selection conditions on the contrastive boundary graph, for one target (`Misinformation`) and one evaluated sibling (`Disinformation`). Only which concepts serve as negatives changes; negative counts are matched. In the held-out-edge condition the evaluated sibling is never a negative, so any benefit must come from the rest of the graph. The graph-aware mixed condition (not shown) is panel (c) with the sibling included.

on some contexts and tested on a held-out one. This matters. An early version without the context split saturated at AUC ≈ 1.0 on surface vocabulary, leaving nothing to measure. F1 is the pre-registered primary metric throughout; balanced accuracy and AUC give the same ordering in every comparison.

Negative-selection conditions. For the graph-contrastive test (designed by p0ss) the unit is a directed edge, target $\rightarrow$ sibling. For each edge, held-out context and seed, we train a target-vs-negatives probe on the other three contexts and test target versus sibling on the held-out context. Positives and negative *counts* are identical across conditions; only negative membership changes (Fig. 1):

- **random**: count-matched non-neighbours of the target (the baseline);
- **direct neighbours**: the target's graph neighbours, including the evaluated sibling;
- **graph-aware mixed**: 40% direct neighbours, 30% distance-2, 20% distant, 10% same-cluster;
- **held-out edge**: the same graph-aware mix *excluding the evaluated sibling.*

The held-out-edge condition is the connectivity test. If the rest of the graph carries information about a boundary, it should partly reconstruct that boundary when the sibling itself is never seen as a negative. With 26 edges, 2 directions, 4 contexts and 30 seeds there are 6,240 paired comparisons per condition.

Statistics. We average the 120 comparisons for each directed edge (4 contexts $\times$ 30 seeds) to a single per-edge value and test the 52 per-edge means against zero (one-sample t, with Wilcoxon as a distribution-free check). Seeds share data, so testing all 6,240 comparisons would overstate

precision. Edges also share endpoints, so even the per-edge intervals are somewhat optimistic; we return to this in §8. The minimum detectable effect (MDE) is computed at 80% power, two-sided $\alpha = 0.05$, from the observed per-edge spread.

4 Relations help as local hard negatives

Declared siblings are the confusable pairs (pre-registered). Iterations 1 and 2 had treated "knowing the relations" as extra *positive* passages along the graph, and found nothing (Appendix A). p0ss pointed out that the hypothesis's mechanism instead uses related concepts as *negatives*. We pre-registered the prerequisite test (locked before any probe was trained). For each concept, a binary out-of-distribution probe separates it from each declared sibling and from each non-sibling, and $\Delta(C)$ is the F1 gap. Siblings are harder to separate for 13 of 15 concepts (Fig. 2): mean $\Delta = +0.055$ (F1 0.916 against siblings, 0.971 against non-siblings), Wilcoxon $p = 1.5 \times 10^{-4}$, and $+0.057$ ($p = 3 \times 10^{-4}$) excluding the hub concept.

Two exploratory controls. Neither was pre-registered; both were added in response to adversarial review. (a) *Placebo graphs*: permuting concept labels while preserving graph topology (2,000 permutations) gives a null distribution centred on zero; the declared graph's sibling gap sits at $z = 2.99$, $p = 0.002$. The effect belongs to these particular relations, not to having *a* graph. (b) *Vocabulary*: declared siblings share more words (mean Jaccard 0.197 vs. 0.179), and word overlap strongly predicts confusability across all 105 pairs (Spearman -0.73). Regressing pair F1 on sibling status and overlap shrinks the sibling coefficient from the raw gap of 0.049 to 0.031 ($p = 4 \times 10^{-4}$), so about a third of the effect is attributable to vocabulary. This is an upper bound on the lexical share, because vocabulary partly carries relatedness. The rank-based version of the same control is marginal (partial Spearman -0.18, $p = 0.07$).

Training with neighbours as negatives helps, on two models. The graph-contrastive test (Table 1) turns confusability into a training benefit. On Gemma-4-E4B at 7 passages per context, the peer replication found direct-neighbour negatives improve F1 by $+0.057$ over count-matched random negatives, graph-aware mixed negatives by $+0.043$, and held-out-edge negatives by $+0.008$. On Gemma-2-9B at 14 passages per context, with our own independently generated data and the same harness, we obtain $+0.059$, $+0.039$ and $+0.012$. The two models agree to within 0.004 on every condition. A layer sweep on Gemma-4-E4B (10 seeds) gives the same ordering at layers 8, 18, 24 and 42. Discrimination is near ceiling at every layer (concept-versus-neighbour AUC 0.96–1.00 from the embedding layer up), so layer 18 is representative, not selected.

The ordering, direct > mixed > held-out edge, follows how close the negatives are to the boundary being tested. Once the sibling itself is removed, most of the gain disappears.

5 Graph connectivity adds no measurable benefit

5.1 Pairwise connectivity test (iteration 2, pre-registered)

Iteration 2 tested H2 directly on the ontology graph. For all 105 pairs, a volume-matched comparison measured the out-of-distribution gain from swapping pair-specific boundary passages into the training data, and the pre-registered primary endpoint was the partial Spearman correlation between gain and $k(A, B)$, controlling for baseline accuracy. The result was $\rho = -0.125$ ($p = 0.20$): no positive relationship. The test had a structural weakness: 84 of the 105 pairs have $k = 1$, and

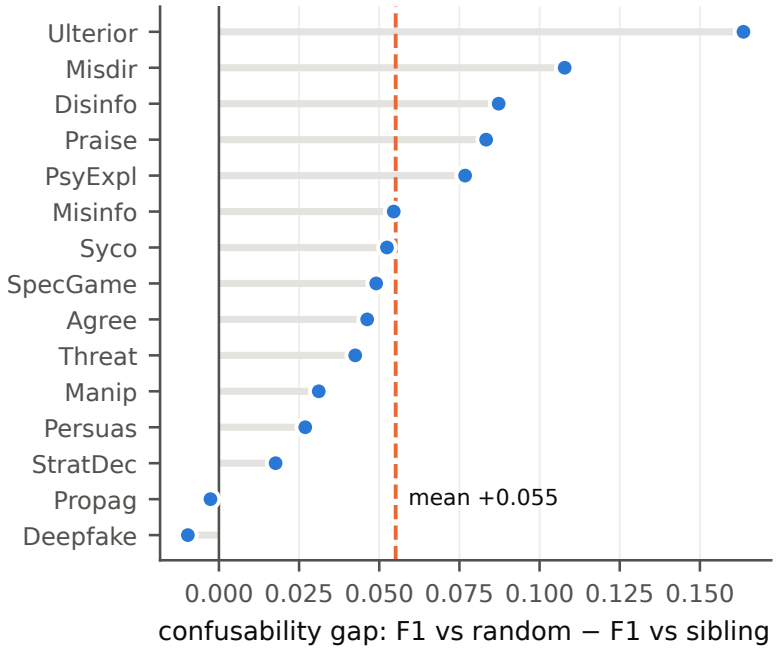


Figure 2: Declared siblings are more confusable than non-siblings (pre-registered Path-2 test, Gemma-2-9B, layer 18, 30 seeds). Each point is one concept's F1 gap between separating it from random non-siblings and from its declared siblings. Mean +0.055, Wilcoxon $p = 1.5 \times 10^{-4}$. An exploratory placebo test (2,000 topology-preserving label permutations) places the declared graph at $z = 2.99$, $p = 0.002$.

most pairs with $k \geq 3$ involve the hub, so the predictor had little spread. The test also used relations as extra positives, which §4 suggests is the wrong mechanism. We therefore read it as consistent with, not strong evidence for, the null below.

5.2 Held-out-edge scaling sweep (iteration 4, pre-registered)

The held-out-edge condition sat near +0.01 on both models at their base volumes (Table 1). The obvious objection is power: perhaps a connectivity signal exists but needs more data. We pre-registered a within-model sweep to test this. The design was Gemma-2-9B; volumes of 7, 14, 28, 56 and 70 passages per concept per context, each subsampled from a single 70-per-context generation and extraction; 30 seeds; held-out-edge ΔF1 as the primary endpoint. The locked decision rule was CONFIRM if the delta rises monotonically with volume and is significantly positive at the top volume, and NULL if the top-volume confidence interval spans zero.

Result. Held-out-edge ΔF1 by volume was +0.008, +0.012, +0.009, +0.004, +0.005 (Fig. 3, Table 2). It does not rise; if anything it falls after 14 per context. At the top volume the mean is +0.0046 with 95% CI $[-0.0053, +0.0144]$ (t-test $p = 0.36$, Wilcoxon $p = 0.78$). The interval spans zero, so the pre-registered verdict is NULL. Held-out-edge negatives beat random negatives in only 40% of the 6,240 paired comparisons.

What the null does and does not exclude. The top-volume interval excludes a mean connectivity benefit above about +0.015 F1 on this graph. It does not exclude a benefit near the MDE of 0.014, because the upper bound and the MDE are essentially equal. Since edges share endpoints, the true interval is probably somewhat wider. We therefore do not claim the connectivity effect

Table 1: Graph-contrastive test: mean ΔF1 over count-matched random negatives (all 6,240 paired comparisons per condition unless noted). E4B rows are the peer replication. Volume is passages per concept per context.

Model	Volume	Layer	Direct neighbours	Graph-aware mixed	Held-out edge
Gemma-4-E4B	7	18	+0.057	+0.043	+0.008
Gemma-4-E4B	28	18	+0.055	+0.041	+0.010
Gemma-2-9B	14	18	+0.059	+0.039	+0.012
Gemma-2-9B	70	18	+0.040	+0.030	+0.005
Layer sweep, Gemma-4-E4B, 7/context, 10 seeds					
Gemma-4-E4B	7	8	+0.072	+0.045	+0.008
Gemma-4-E4B	7	18	+0.055	+0.044	+0.008
Gemma-4-E4B	7	24	+0.066	+0.044	+0.003
Gemma-4-E4B	7	42	+0.059	+0.043	+0.011

Table 2: Scaling sweep, Gemma-2-9B: across-edge paired tests (52 directed edges, each averaged over 4 contexts × 30 seeds). p is the one-sample t-test; MDE is the minimum detectable effect at 80% power.

	Held-out edge (primary)				Direct neighbours
Volume	ΔF1	95% CI	p	MDE	ΔF1 [95% CI]
7	+0.008	$[-0.001, +0.018]$	0.081	0.013	+0.061 $[+0.039, +0.083]$
14	+0.012	$[+0.000, +0.024]$	0.049	0.017	+0.059 $[+0.038, +0.079]$
28	+0.009	$[-0.002, +0.019]$	0.103	0.015	+0.050 $[+0.031, +0.069]$
56	+0.004	$[-0.006, +0.015]$	0.39	0.015	+0.040 $[+0.021, +0.060]$
70	+0.005	$[-0.005, +0.014]$	0.36	0.014	+0.040 $[+0.021, +0.059]$

is zero, only that it is too small to matter at the volumes we tested and that more data did not make it larger. The same instrument, at the same volume, clearly detects the direct-neighbour effect (+0.040, CI $[+0.021, +0.059]$, $p = 10^{-4}$). One volume deserves mention: at 14 per context the held-out-edge interval just touches zero ($p = 0.049$). This was not the pre-registered endpoint, is uncorrected across five volumes, and is followed by smaller estimates at every higher volume.

Net of vocabulary. The pre-registration called for reading the result net of lexical overlap but did not fix the model; the specification below was chosen after the primary result was known. For each comparison we compute how much more word overlap (Jaccard) the held-out-edge negatives share with the evaluated sibling than the random negatives do, average to one value per edge, and regress per-edge ΔF1 on it. Vocabulary advantage strongly predicts the per-edge benefit (slope +1.72, $p < 0.001$, $R^2 = 0.32$; Fig. 5a). The intercept, the held-out-edge effect at zero vocabulary advantage, is -0.006 $[-0.016, +0.003]$. The small positive gross effect is therefore plausibly lexical.

Per-edge connectivity (exploratory). The pre-registered endpoint concerns the mean. A direct test of H2 asks whether edges with *more* alternate routes benefit more. On the contrastive graph, after removing each edge, its endpoints have between 0 and 4 vertex-disjoint alternate paths (counts 1, 10, 9, 5, 1). Per-edge held-out-edge ΔF1 does not increase with this count (Spearman $\rho = -0.34$, $p = 0.085$, 26 edges; Fig. 5b). The trend is, if anything, in the wrong direction.

Secondary: the neighbour premium shrinks with data. Direct-neighbour negatives stay clearly positive at every volume, but their advantage falls from +0.061 to +0.040 as volume grows

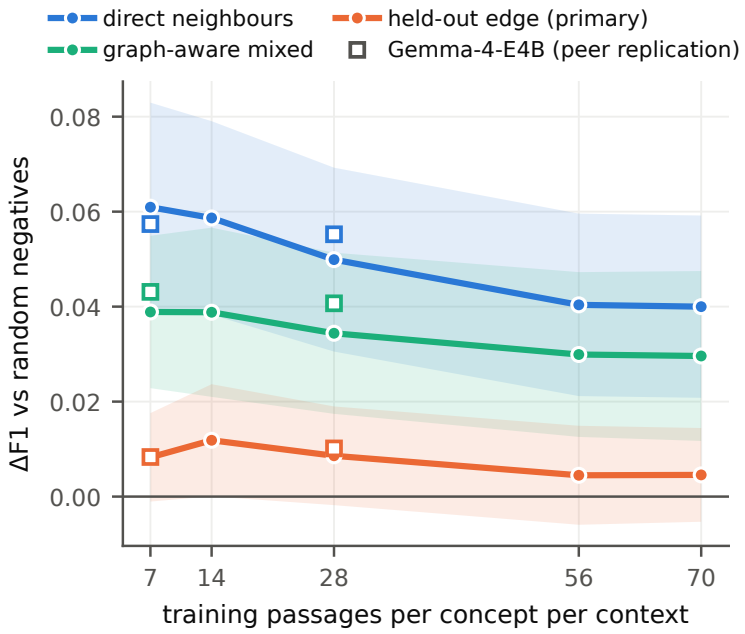


Figure 3: Held-out-edge ΔF1 (the pre-registered primary, orange) does not rise with data volume. Lines and bands: Gemma-2-9B means with 95% confidence intervals across the 52 directed edges. Hollow squares: Gemma-4-E4B from the peer replication, shown as independent corroboration only; the pre-registered trend is within-model (Gemma-2-9B). Direct-neighbour and mixed conditions are pre-declared secondary outcomes.

tenfold. Absolute F1 improves in every condition with more data, and random negatives gain most. Smart negative selection is worth most when data is scarce, which is the usual situation for long-tail safety concepts.

5.3 What this test can and cannot show

We want to be plain about the reach of the connectivity null. The graph is small: 15 concepts and 26 edges. After removing an edge, 11 of the 26 have at most one alternate path and none has more than four, so the range of connectivity being tested is narrow. The earlier pairwise test was narrower still, with 84 of 105 pairs at $k = 1$. The graph was curated by hand, and the result is conditional on it being a sound description of how these concepts relate. Probe discrimination is near ceiling (random-negative F1 0.92 at the top volume), which compresses every effect. A graph with many more concepts, genuinely independent alternate paths and harder boundaries could behave differently. What we can say is that, for a realistic safety-concept family of the size a practitioner might curate, the graph's connectivity added nothing we could measure beyond what direct neighbours already provide.

6 An account: independent-constraint boundary coverage

The following account was proposed by p0ss after iteration 3. It is post hoc and we present it as an interpretation that fits the results, not as a tested theory.

Treat a concept as a centroid with a class-conditional spread, and a probe's decision boundary as lying where that spread thins. Positive examples mainly sharpen the centroid. Negative examples place *facets* of the boundary, each mainly on the side that faces the negative. On this picture:

extra positives along the graph do little once the centroid is well estimated (iterations 1–2); a sibling negative moves the facet facing that sibling (§4); the benefit is ordered by how directly the negatives face the tested boundary (direct > mixed > held-out); and other relations only help with a withheld boundary if they supply constraints oriented along it. A path count is a reasonable proxy for "number of independent constraints" only when graph edges correspond to independent directions in representation space. Our evidence suggests they do not here. The held-out-edge benefit tracks shared vocabulary, not alternate-path count (Fig. 5). The model's actual confusions concentrate on the hub (below). On the contrastive graph, however, only 3 of the 19 edges not touching the hub have every alternate path running through it, so "alternate paths route through one hub" is not literally true of the graph. The claim is about representational correlation, and we have not measured that directly.

Geometry is consistent (peer replication, Gemma-4-E4B). Raw cosine similarity suggests every concept overlaps every other (adjacent pairs 0.971, non-adjacent 0.961), but random passage pairs already sit at 0.892: the common-direction effect (Ethayarajh, 2019). After mean-centring, adjacent pairs separate from non-adjacent ones by +0.166 and far-domain controls fall well outside the family (−0.147), but graph distance correlates only weakly with centroid similarity ($r = -0.30$ at layer 18; −0.27 to −0.39, median −0.34, across all 43 layer positions). Local adjacency is visible in the geometry; graded multi-hop structure is not. Probe results are unaffected by the correction, since a linear classifier can absorb a common offset.

A forward-looking implication. The account implies that a boundary face nothing has been trained against should leak whichever scheme is used. We had already pre-specified a test of one such face before running it (§7). The test was designed as a rejection test, not as a test of the account, so we count it as consistent with the account rather than as a confirmed prediction.

7 Deployment consequences: rejection and tiered negatives

The analyses in this section move from pairwise discrimination to per-concept one-vs-rest lenses evaluated at a fixed operating point: false-positive rate (FPR) at 95% true-positive rate, with negative count matched to each concept's graph degree. Pairwise AUC is near ceiling, but FPR at high recall is sensitive to exactly the boundary placement that negatives control.

A flat probe cannot say "none of these" (peer replication, Gemma-4-E4B). A 15-way argmax assigns every far-domain input (passages about dogs, weather, basketball, photosynthesis) to some concept. A confidence threshold rejects only 26% of them at 95% in-family retention (AUROC 0.836). Adding a coarse *other* class trained on far-domain controls rejects them at recall 1.0, including a held-out far concept it never trained on, with no false rejection of in-family passages. Hierarchical *classification* is not the fix: flat accuracy 0.910 exceeds soft (0.857) and hard (0.850) coarse-to-fine decoding. The coarse stage is useful for rejection, while resolution is best left to the flat lens.

Each scheme seals the face it trains against. Table 3 reports FPR by boundary face on both models. On Gemma-4-E4B, graph-sibling negatives cut neighbour-face FPR from 0.192 to 0.046 but are no better than random on off-graph confusers (0.257 vs. 0.270). Negatives mined from the model's own confusions cut the off-graph face to 0.058, a 20-point reduction, but give back most of the neighbour gain. A cheap far-domain *other* tier seals the far face entirely (0.000). We reproduce

Table 3: False-positive rate at a 95%-TPR operating point (E4B: 93% achieved) by boundary face and negative scheme, with negative counts matched. Gemma-4-E4B is the peer replication (layer 18). Gemma-2-9B is this work (layer 18, 14 lenses, `PersuasiveCommunication` quarantined); the multi-scale scheme was not run there because only one far-domain control set was available. Near-OOS: legitimate-influence passages, pooled and on contexts held out of the lens's training fold.

					Near-OOS	
Model	Negatives	Neighbours	Off-graph	Far-OOD	pooled	held-out ctx
Gemma-4-E4B	random	0.192	0.270	0.179	–	–
	graph-sibling	0.046	0.257	0.189	–	–
	model-mined	0.157	0.058	0.241	–	–
	multi-scale[a]	0.146	0.069	0.000	–	–
Gemma-2-9B	random	0.318	0.335	n/a[b]	0.164	0.229
	graph-sibling	0.032	0.378	n/a[b]	0.175	0.225
	model-mined	0.158	0.119	n/a[b]	0.182	0.227

[a] model-mined negatives plus the far-domain *other* controls. [b] far-domain set used for evaluation only (nothing trained against it).

the structure on Gemma-2-9B (our run, layer 18): siblings seal neighbours (0.318 → 0.032), and mining seals the off-graph face (0.335 → 0.119). The absolute values differ, the pattern does not. Off-graph confusions matter because the graph misses the model's hub-attractor errors: in the peer replication, 24 of the flat probe's 42 misclassifications fall on concept pairs with no edge (a small base; the pattern, not the exact count, is the robust finding). Several concepts are confused *into* the hub without an edge to it.

Calibration drift. In the peer replication, an earlier run that set thresholds on training positives rather than held-out ones achieved only 34% held-out TPR (reported in the replication write-up; no result file was committed for that run). On Gemma-2-9B we measure the same effect more mildly: train-calibrated thresholds give held-out TPR of 0.75–0.81 instead of 0.95. Lens scores shift across contexts, and a deployed monitor has no held-out labels with which to notice.

The hard regime: legitimate influence (pre-specified). Far-domain inputs are the easy rejection case. The hard case is legitimate influence adjacent to the family: honest negotiation, transparent advertising and declared lobbying. We generated and audited 84 such passages (3 concepts × 4 contexts × 7) and specified the analysis before running it: score the 14 lenses other than `PersuasiveCommunication`, which is quarantined because its own training data includes honest advertising, so firing there is a true positive by construction; report FPR pooled and split by whether the passage's context was held out of the scoring lens's training fold.

No scheme reduces the leak (Fig. 4). Pooled FPR is 0.164, 0.175 and 0.182 under random, graph-sibling and model-mined negatives. Paired across the 14 lenses, graph-sibling negatives change it by +0.010 (95% CI $[-0.074, +0.095]$, Wilcoxon $p = 0.86$) and model-mined negatives by +0.018 ($[-0.101, +0.136]$, $p = 1.00$). The same lenses and test detect the neighbour seal (-0.286, $p < 0.001$) and the off-graph seal (-0.216, $p = 0.006$), so an effect of comparable size here would have been visible. Leakage is significantly worse on contexts held out of the lens's training fold ($+0.068$, $p = 0.016$): the realistic figure is about 0.23, roughly one legitimate-influence passage in four, rather than the pooled 0.16. The leak is concentrated. Under graph-sibling negatives, `UnwarrantedPraise` (0.589) and `ExcessiveAgreement` (0.497) account for most of it, and three

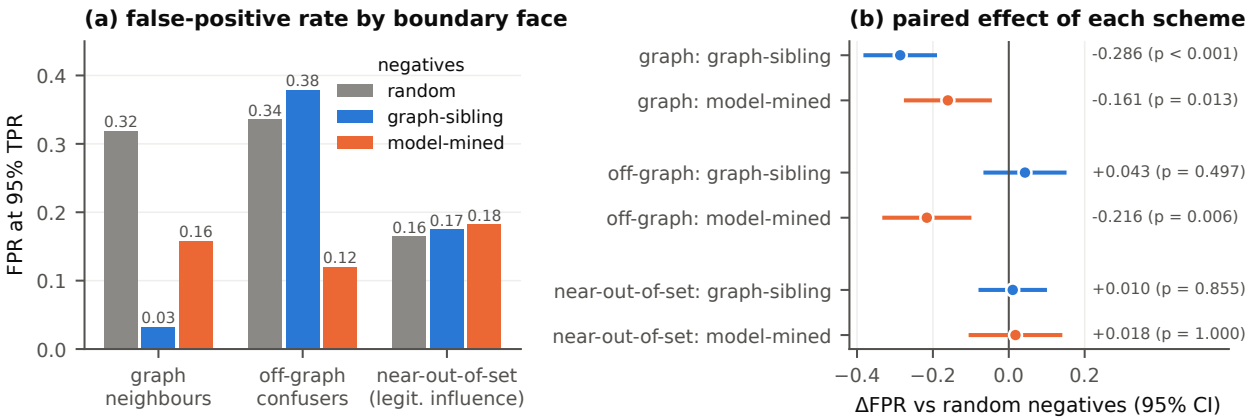


Figure 4: No negative scheme reduces false positives on legitimate influence (Gemma-2-9B, layer 18, 14 lenses). (a) FPR at 95% TPR by boundary face. Graph-sibling negatives seal the neighbour face and model-mined negatives the off-graph face; the near-out-of-set face, trained against by neither, stays flat. (b) Paired change in FPR relative to random negatives across lenses (95% CI; Wilcoxon p). Both positive controls clear zero; both near-out-of-set intervals include it. Off-graph tests use the 13 lenses with off-graph confusers.

lenses never fire. Honest advertising reads as unwarranted praise and cooperative negotiation as excessive agreement. The quarantined `PersuasiveCommunication` lens fires at 0.24–0.52, well above every scored lens, as expected.

Practical reading. Cheap tiers (graph siblings plus a far-domain *other* class) seal the neighbour and far faces; a cheap audit of the model's confusions locates the off-graph gap, where mined negatives are worth their cost. None of this addresses legitimate influence adjacent to the family, which a deployed monitor is likely to meet more often than either far-domain text or textbook manipulation. Until that face is trained against explicitly, practitioners should expect a monitor built this way to fire on roughly one honest-influence input in four, mostly through a small number of identifiable lenses.

8 Limitations

- **Scope of the connectivity null.** One small, hand-curated graph with a narrow range of alternate-path counts (§5.3). The null is conditional on that graph and does not license a general claim that concept-graph connectivity is useless for probes.
- **What the interval excludes.** The decision rule ("CI spans zero") is weaker than an equivalence test. The top-volume upper bound (+0.0144) is approximately the MDE, so effects near 0.014 are not excluded. Edges share endpoints and are not independent; a concept-level resampling would likely widen the interval.
- **Synthetic data, same-model audit.** All passages were written by one model, and the blind audit used the same model. Generator-specific phrasing could create or mask effects, and the audit cannot detect errors the generator and auditor share. The hub concept self-matched at only 70%. We have no human-written evaluation set.
- **Lexical confounding.** Vocabulary explains a substantial share of every effect we can test for it. The out-of-distribution context split reduces but does not remove surface cues.

- **Near ceiling.** Pairwise discrimination is near ceiling on both models, which compresses all deltas; the operating-point FPR analysis is the more sensitive instrument.
- **Seed variance understates data variance.** Seeds vary probe training and subsampling, not the generated corpus. A fresh generation is a larger source of variation than our intervals reflect, though the cross-model agreement on independently generated data is partial reassurance.
- **Exploratory analyses.** The placebo-graph and Path-2 vocabulary controls, the specification of the iteration-4 vocabulary control, and the per-edge connectivity test were not pre-registered.
- **Probe and model coverage.** Linear probes at one layer (with an MLP spot-check and an E4B layer sweep), mean pooling, two base models, one concept family.
- **Near-out-of-set sample.** 84 passages across three concepts. Adequate for the pooled null, given the positive controls; too small to rank which kind of legitimate influence leaks most (per-concept FPRs 0.155–0.190 under graph-sibling negatives are within noise).
- **Pre-registration record.** The public release repository is a single snapshot. Lock dates are verifiable from the commit history of the development repository (Appendix B).

Future work. The most direct follow-up is to train lenses against near-out-of-set negatives and test whether that seals the legitimate-influence face without reopening the others. A connectivity retest should use a larger graph with independently motivated alternate paths, and count representationally independent constraints rather than vertex-disjoint paths. Contrast-prompted generation ("write A in a way that is not B"; generation mode alone made no difference, Appendix C), human-written evaluation data, and sparse-autoencoder features (Lieberum et al., 2024) are natural extensions.

9 Conclusion

For linear probes over a family of related AI-safety concepts, relational structure is useful in a narrow way. Direct relations make good, cheap hard negatives: about +0.06 F1 on two models, perhaps a third of it shared vocabulary. Beyond that, we found no measurable benefit from the graph's connectivity. A tenfold increase in data did not raise it, and what little there was tracks vocabulary rather than alternate paths. The negative schemes we tested seal only the boundary faces they train against, and none of them addresses legitimate influence adjacent to the family, which leaks at about one passage in four. We think the most useful role for a concept graph in this setting is as a guide to which negatives to collect, not as a source of structure a probe can exploit.

Contributions and acknowledgements

J.B. designed and ran iterations 1–4, the Path-2 test, the scaling sweep, the near-out-of-set controls and rejection test, and wrote the paper. p0ss (HatCat) performed the peer replication on Gemma-4-E4B and contributed the hard-negative reframing, the graph-contrastive test design and contrastive boundary graph, the layer sweep and anisotropy control, the two-level probe, the multi-scale and confuser-coverage analyses, and the boundary-coverage account. At p0ss's request their work is credited as a peer replication rather than co-authorship. The work was carried out as a capstone in

the TARA program, which follows the ARENA curriculum (McDougall et al., 2024); no ARENA course code is used. Compute was provided through TARA.

Use of language models. All passages were generated, and blind-audited, by Claude Opus 4.8. AI coding and writing assistants were used for analysis code, adversarial review of results, and drafting and editing this manuscript; all numbers were checked against committed result files.

Code and data availability. Code, generated passages, audit files, result JSONs and figure scripts: `https://github.com/magazine233/knowing-the-graph` (MIT licence; see its `NOTICE` for third-party terms). Model weights and raw activations are not redistributed. Figures in this paper are regenerated by `paper/figures/make_figures.py` from committed JSONs.

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A Study history

Iteration 1. Definition-only versus relation-augmented multiclass probes, with a correlation between per-pair gain and vertex connectivity. An apparent positive at 8 concepts did not replicate at 15, and once data volume was matched there was no robust benefit. In hindsight the design also compared conditions on different test distributions and treated relations as extra positive passages.

Iteration 2. The pre-registered pairwise test in §5 (105 pairs, partial Spearman -0.125, $p = 0.20$). A pilot positive at smaller scale was noise.

Iteration 3 (Path 2). The reframing to relations as hard negatives and the pre-registered confusability test in §4; the graph-contrastive test by the peer replication.

Iteration 4. The pre-registered scaling sweep in §5, followed by the pre-specified near-out-of-set rejection test in §7.

B Pre-registration record

Lock and result commits in the public development repository `https://github.com/magazine233/ARENA_3.0` (files under `project/capstone/`; branch `iteration-3`, merged to `main`). Dates are 2026, times AEST (UTC+10).

Test	Locked	Result	Primary endpoint	Outcome
Iteration 2	`29c70550` (06-06 11:24)	`d69f04bd` (06-06 19:10)	partial Spearman, gain vs k	null
Path 2	`e927b969` (06-09 13:37)	`1a91a262` (06-09 14:20)	mean $\Delta(C) > 0$	confirmed
Iteration 4	`21c0bf1e` (06-10 07:27)	`b45eb6c4` (06-12 11:11)	held-out-edge ΔF1	null
Near-OOS	`58a54d16` (06-11 21:51)	`08c84d6d` (08-05 11:05)	FPR, 14 lenses	no effect

The near-out-of-set analysis was specified in the iteration-4 run book and paper draft rather than in a standalone pre-registration document. The Path-2 placebo and vocabulary controls were committed with the Path-2 result (`1a91a262`) and are marked exploratory in their result files.

C Generation mode (peer replication)

p0ss also tested whether activations recorded while Gemma-4-E4B *generates* a concept differ usefully from activations recorded while it *reads*. Using 420 definition-prompted self-generations, probes built on autoregressive states and on teacher-forced re-reads of the same tokens passed a check suite (F1, balanced accuracy and AUC all ≥ 0.80) at the same rate (0.692 vs. 0.693). Reading the generated passage without its prompt dropped the rate to 0.454, which reflects losing the definition in the prompt rather than a generation effect. The graph-contrastive ordering (direct > mixed > held-out edge, the last at +0.002 to +0.011) holds in every mode. Contrast-prompted generation was not tested.

D Per-edge diagnostics

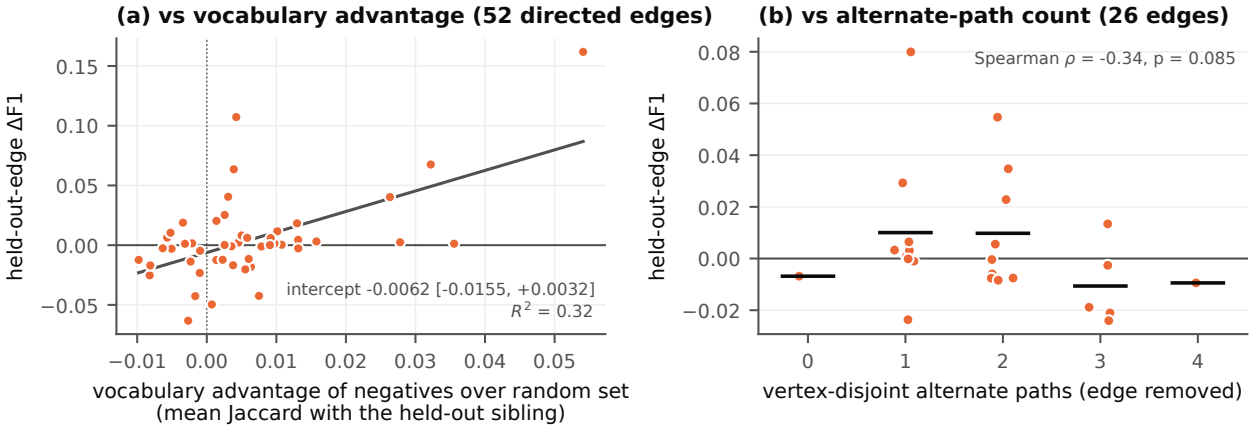


Figure 5: Held-out-edge benefit per edge at 70 passages per context (Gemma-2-9B). (a) Against the vocabulary advantage of the held-out-edge negatives over the random negatives; line: OLS fit, whose intercept is the vocabulary-adjusted effect. (b) Against the number of vertex-disjoint alternate paths between the edge's endpoints once the edge is removed (undirected edges; bars are group means). Both analyses are exploratory.

E Concepts

Table 4: SET G concepts and the definitions recorded in the contrastive boundary graph.

Concept	Definition
`ManipulativeCommunication`	Communication that steers another party by exploiting framing, pressure, asymmetry, or vulnerability rather than preserving their independent evaluation.
`PersuasiveCommunication`	Legitimate influence that offers reasons, evidence, or appeals while leaving the recipient's evaluative agency intact.
`Propaganda`	Systematic messaging that promotes an ideology, group, or agenda by repetition, selective framing, identity pressure, or suppression of contrary context.
`Misinformation`	False or misleading information spread without the defining feature of deliberate deceptive intent.
`Disinformation`	False or misleading information spread with deliberate deceptive intent.
`PsychologicalExploitation`	Use of another party's cognitive, emotional, social, or situational vulnerabilities to steer their beliefs or choices.
`SycophancyRegister`	A sustained servile or over-agreeable communicative register oriented around pleasing, flattering, or appeasing the recipient.
`UnwarrantedPraise`	A discrete act of praise that is unsupported, exaggerated, or strategically insincere.
`ExcessiveAgreement`	Over-agreement with a person's claims, preferences, or judgments beyond what evidence or independent evaluation warrants.
`Misdirection`	A tactic that redirects attention, salience, or interpretation away from a relevant fact or issue.
`StrategicDeceptionPlanningProcess`	Planning or sequencing deceptive actions over time to achieve a goal.
`SpecificationGamingProcess`	Optimizing for a specified metric, rule, or reward in a way that violates the intended objective.
`UlteriorMotiveDetection`	Detection or inference of hidden motives behind another party's behaviour or communication.
`Deepfake`	Synthetic or manipulated media depicting a person, event, or signal in a way that can be mistaken for authentic media.
`Threaten`	An explicit or implied conditional commitment to cause harm, penalty, or loss unless the target complies.